

DRAFT — DBA-GAS-FC2: Natural Gas Processing

Natural Gas Processing

Design Basis Accident — Military Action

DBA-GAS-FC2 — Facility Class Document

Issued Under: DBA-GAS Sector Framework

Version 0.1 — DRAFT

June 2026

Timothy E. Roxey

Eclectic Technologies

Prepared for: Natural Gas Sector — Industry, Government, and Allied Partners

Parent document: DBA-GAS Sector Framework v1.4

Preface

The natural gas processing plant occupies the most analytically underappreciated position in the natural gas supply chain. It sits between the wellhead and the transmission system, visible to neither the pipeline engineer nor the end-user, performing the unglamorous but irreplaceable function of separating what the transmission system cannot accept — natural gas liquids, water, hydrogen sulfide, carbon dioxide — from what it can. When a major processing plant is destroyed, upstream production has nowhere to go: wellheads must be shut in, not because the wells are damaged, but because the gas they produce cannot be brought to pipeline quality. The disruption propagates upstream and downstream simultaneously, and it does so silently — without the visible fire of a refinery attack or the measurable pressure drop of a pipeline rupture.

This document establishes the Design Basis Accident scenarios for deliberate military action against natural gas processing infrastructure. The facility class encompasses gas processing plants that perform NGL extraction, acid gas removal, dehydration, and fractionation — the full range of processing operations that transform raw wellhead gas into pipeline-quality dry gas and marketable NGL products. The primary reference environments are the Permian Basin, the Midcontinent, and the Gulf Coast processing corridors, where processing plant concentration creates the same cluster vulnerability that characterizes the Gulf Coast refinery complex in the petroleum sector.

The distinguishing analytical contribution of FC2 is the dual consequence chain. Every other facility class in the DBA-GAS series produces a single primary consequence: transmission disruption, storage loss, or distribution outage. A processing plant attack produces two simultaneous and independent consequence chains — gas supply curtailment that propagates through the transmission system to end-users, and NGL supply curtailment that propagates through the petrochemical feedstock system to plastics, fertilizers, and chemical manufacturing. The adversary who understands both consequence chains can select processing plants whose loss maximizes compound disruption across two sectors simultaneously.

FC2 also carries the most consequential AI governance challenge in the DBA-GAS series. Processing plants are among the most heavily optimized continuous-process facilities in the energy sector — APC and real-time optimization systems continuously adjust process conditions to maximize NGL recovery at current market prices. This optimization infrastructure shares its architecture with the safety-critical DCS that protects against overpressure, cryogenic release, and fired heater excursion. The Bright Line between commercial optimization and process safety runs through the middle of the same control system, and the commercial pressure to maximize liquid recovery creates a continuous incentive to push process conditions toward the margins where safety systems are most likely to be tested. This is not an adversarial risk alone. It is a structural risk present in every processing plant that has deployed advanced process control without architectural separation of the optimization layer from the safety layer — and it is the

risk that makes processing plants particularly attractive targets for adversaries who understand the control plane.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the natural gas processing facility class. Its purpose is to define the bounding threat event that processing plant security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A processing plant security program designed against single-vector physical threats will fail when a sophisticated adversary combines DCS pre-positioning — exploiting the same IT-OT boundary vulnerabilities present in every modern processing plant — with precision kinetic action against the process units whose loss simultaneously maximizes gas supply disruption and NGL production curtailment. The DBA-MA approach defines the coordinated, multi-vector attack as the governing design event, with the AI governance interaction as both an attack pathway and an independent structural risk that the Design Basis must address.

1.2 Scope and Applicability

This document addresses natural gas processing facilities: plants that receive raw wellhead gas and deliver pipeline-quality dry gas and NGL products through one or more of the following operations: NGL extraction by Joule-Thomson expansion, turbo-expander, or lean oil absorption; acid gas removal by amine treating or membrane separation; dehydration by glycol absorption or molecular sieve; and NGL fractionation, separating the mixed NGL stream into purity products (ethane, propane, isobutane, normal butane, natural gasoline). The facility class encompasses both gathering-system processing plants in producing fields and straddle plants on transmission systems.

Natural gas transmission pipelines and compressor stations are addressed in DBA-GAS-FC1. Underground storage is addressed in DBA-GAS-FC3. Local distribution is addressed in DBA-GAS-FC4. LNG terminals are addressed in DBA-GAS-UC-LNG. Where processing plant events produce consequences that propagate into the transmission system, this document addresses those consequences as they originate at the processing plant without displacing the DBA-GAS-FC1 analysis of the transmission consequence. The NGL consequence chain — from processing plant through fractionation to petrochemical feedstock — is within the scope of this document; the downstream petrochemical facility consequences are noted as Tier 3 cross-sector cascades without displacing dedicated analysis in the relevant chemical sector instruments.

1.3 Relationship to Governing Documents

This document is issued under the DBA-GAS Sector Framework (v1.4), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the processing-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier consequence framework, and the adequacy standards are inherited unchanged from the sector framework. The bounding event categories are inherited with processing-specific adaptation. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the processing plant control environment in Section 3.3, with the additional recognition that the non-adversarial AI risk in processing plants requires the same architectural constraints as the adversarial risk — a distinction unique to this facility class within the DBA-GAS series.

1.4 The Processing Plant as the Dual Consequence Node

A fundamental characteristic of the natural gas processing facility class distinguishes it from every other facility class in the DBA-GAS series: a processing plant attack simultaneously disrupts two independent and economically significant supply chains. The gas supply disruption propagates downstream through the transmission system to end-users — power generators, industrial consumers, and residential heating customers — in the same manner as a compressor station outage. The NGL supply disruption propagates through a separate supply chain into the petrochemical feedstock system, curtailing the ethane supply that feeds ethylene crackers, the propane supply that feeds residential heating and agricultural drying, and the butane supply that feeds gasoline blending and chemical manufacturing.

These two consequence chains operate independently and simultaneously. A processing plant that is shut down for gas supply reasons stops NGL production regardless of the status of the downstream fractionation system. A fractionation unit that is destroyed continues to constrain NGL product delivery even after the upstream processing plant resumes operations. The adversary who targets a processing complex can maximize compound disruption by selecting targets that sever both chains at their most constrained points — and the intelligence required to make that selection is available in public-domain FERC filings, EPA RMP submissions, and pipeline flow data.

2. Processing Plant Profile and Asset Taxonomy

2.1 Processing Operations and Unit Types

Inlet separation and compression. Raw wellhead gas arrives at the processing plant at variable pressure and temperature, carrying liquids — water, condensate, and NGLs — entrained in the gas stream. Inlet separators remove bulk liquids; inlet compression raises gas pressure to the plant operating level. Inlet compression is often the single most consequential piece of equipment at a processing plant. If the inlet compressor fails, no gas enters the plant regardless of the status of downstream processing units.

NGL extraction. The core processing function. Three primary technologies are in commercial use. Joule-Thomson (JT) expansion is the simplest: gas pressure is reduced across a valve, cooling the stream by the Joule-Thomson effect and condensing NGLs. Turbo-expander plants recover mechanical energy from the pressure drop, achieving deeper NGL extraction and higher thermodynamic efficiency. Lean oil absorption uses a hydrocarbon solvent to selectively absorb NGLs from the gas stream. Turbo-expander plants dominate modern large-scale processing and are capable of ethane recovery rates exceeding 90 percent; their cryogenic operating temperatures — as low as -150°F — create the most severe cold-temperature process hazard in the DBA-GAS series.

Acid gas removal. Raw gas from sour formations contains hydrogen sulfide (H_2S) and carbon dioxide (CO_2) that must be removed before the gas meets pipeline quality specifications. Amine treating units — typically monoethanolamine (MEA), diethanolamine (DEA), or methyldiethanolamine (MDEA) systems — selectively absorb acid gases from the process stream and regenerate the solvent in a stripper column. H_2S is regenerated as a concentrated acid gas stream that is either incinerated in a thermal oxidizer or fed to a sulfur recovery unit. H_2S handling creates the acute toxic hazard specific to sour gas processing plants — a release of concentrated H_2S from a failed amine unit or thermal oxidizer can produce a lethal vapor cloud extending beyond the plant boundary, with off-site consequence potential analogous to HF alkylation unit releases in the petroleum sector.

Dehydration. Water vapor is removed from the gas stream by glycol absorption (triethylene glycol is most common) or molecular sieve units. Dehydration prevents hydrate formation in downstream cryogenic units and pipelines. Glycol dehydration units are ubiquitous but individually have a small consequence; molecular sieve units in turbo-expander plants have a higher consequence because their failure can force a plant shutdown or a reduction in plant operating temperature limits.

NGL fractionation. The mixed NGL stream from extraction — Y-grade NGL — is fractionated into purity products in a series of distillation columns: the deethanizer separates ethane from the heavier stream; the depropanizer separates propane; the debutanizer separates butane from natural gasoline. Fractionation trains are capital-intensive, have long replacement lead times for major column sections, and represent the highest-consequence physical attack target in the NGL

production chain because their loss stops purity NGL product delivery even when upstream extraction continues.

2.2 Gulf Coast and Midcontinent Processing Clusters

Gas processing capacity in the United States is concentrated in three primary geographic corridors: the Permian Basin / Delaware Basin processing complex in West Texas and New Mexico, the Midcontinent processing complex centered on the STACK, SCOOP, and Anadarko plays in Oklahoma, and the Gulf Coast processing complex associated with offshore production and the onshore Eagle Ford and Haynesville plays. Each corridor exhibits the same cluster vulnerability identified in the petroleum sector: a relatively small number of major processing plants account for a disproportionate share of regional processing capacity, and simultaneous attacks on three or four plants within a corridor can exceed the corridor's total adaptive capacity.

The Permian Basin processing complex is the largest and fastest-growing in the United States, driven by associated gas production from Permian oil wells. A significant fraction of Permian oil production depends on processing plant capacity to handle associated gas; when processing capacity is constrained — as it was during the 2018-2019 processing bottleneck — upstream oil producers are forced to flare associated gas or curtail oil production. An adversary who destroys Permian processing capacity does not merely disrupt gas supply; they can force oil production curtailment from the most prolific crude oil basin in the United States.

2.3 The NGL-Petrochemical Feedstock Connection

NGLs are the primary feedstock for the U.S. petrochemical industry. Ethane from gas processing feeds ethylene crackers — the first step in the production of polyethylene, the most widely produced plastic. U.S. ethylene crackers have restructured over the past decade to run predominantly on ethane rather than naphtha, creating a supply chain dependency on gas processing capacity that has no quick alternative: switching a modern ethane cracker to naphtha feedstock requires months of operational adjustment and a significant yield penalty. Propane from gas processing feeds residential heating, agricultural crop drying, and chemical manufacturing. Butanes feed gasoline blending and alkylation — the same petroleum sector consequence chain identified in DBA-OIL-FC1's HF alkylation section.

The petrochemical consequence of a major processing plant cluster attack is therefore not a secondary or decorative observation. It is a first-order supply chain disruption that affects plastics production, agricultural operations, and military logistics (propane for forward operating base heating and agricultural support) within weeks of the initiating event, without any further attack on downstream petrochemical infrastructure.

2.4 Regulatory Authority

PHMSA. 49 CFR Part 192 governs gas processing plant piping and equipment that meets the definition of a gas pipeline facility. PHMSA's process safety requirements for gathering and processing are less prescriptive than for transmission pipelines; processing plant safety is primarily governed by EPA and OSHA rather than PHMSA for most process equipment.

EPA RMP. The Risk Management Program under 40 CFR Part 68 requires processing plants that handle listed regulated substances — including H₂S, propane, and flammable mixtures — above threshold quantities to prepare worst-case release scenario analyses and emergency response plans. RMP data provides public-domain consequence analysis for H₂S and flammable vapor releases that is directly relevant to the DBA-MA threat analysis.

OSHA PSM. The Process Safety Management standard under 29 CFR 1910.119 requires process hazard analysis, mechanical integrity programs, management of change, and emergency response planning for processing plants handling listed highly hazardous chemicals at or above threshold quantities. PSM is the primary safety regulatory framework for processing plant operations and provides the baseline for the non-adversarial Design Basis.

TSA. TSA Pipeline Security Directives apply to gas pipeline operators and extend to processing plants that are integral to interstate pipeline system operations. Physical security requirements for processing plants remain voluntary, replicating the gap identified across the petroleum and gas sectors.

3. Design Basis: Natural Gas Processing Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the natural gas processing facility class inherits the process safety event, infrastructure failure event, natural hazard event, and operational event categories from the DBA-GAS Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Turbo-expander train fire and explosion. The governing process safety bounding event is a major fire and explosion at a cryogenic turbo-expander train resulting from a high-pressure hydrocarbon release at or near the cold box — the cryogenic heat exchange system operating at temperatures as low as -150°F. Cold-box fires are among the most difficult industrial fires to suppress: the cryogenic process stream continues to feed the fire until isolation is achieved, the low temperatures create ice formations that complicate fire suppression foam application, and the thermal cycling from cryogenic to fire conditions can cause catastrophic mechanical failures in adjacent piping. The governing reference event is the 2017 Midcoast Energy Resources processing plant explosion in Folger, Texas, and the class of Midcontinent processing plant incidents in the 2000s-2010s that established the cold-box fire as the bounding process safety event for turbo-expander facilities. Plant shutdown and cold-box replacement — the governing recovery constraint — requires 6 to 18 months for a major cryogenic train.

H₂S release from sour gas processing. For processing plants handling sour gas, the governing consequence safety bounding event is a large H₂S release from a failed amine unit, thermal oxidizer, or sulfur recovery unit. H₂S is acutely toxic at concentrations above approximately 50 ppm; concentrations above 300 ppm are immediately dangerous to life and health. A large release from a sour gas processing plant can produce a lethal vapor cloud extending multiple miles downwind under stable atmospheric conditions. The EPA RMP worst-case analysis for affected facilities provides the reference consequence envelope. The DBA-MA extension adds the adversarial manipulation pathway — DCS compromise that inhibits the H₂S detection and ESD system response.

Processing bottleneck cascade. The non-adversarial operational bounding event is a regional processing capacity constraint that forces upstream wellhead shut-ins and downstream transmission supply shortfalls simultaneously. The 2018-2019 Permian Basin processing bottleneck — where processing capacity additions lagged production growth, forcing operators to flare associated gas or curtail oil production — is the reference event. The bounding case involves a major processing plant outage during a period of constrained regional spare capacity, producing a supply curtailment that cannot be offset by alternative routing because the processing bottleneck is structural, not locational.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the natural gas processing facility class is a coordinated, multi-vector attack against a major processing plant or cluster of processing plants within a production corridor, combining DCS compromise with precision kinetic action against the highest-consequence process units in both the gas processing and NGL fractionation trains, timed to execute during peak demand season when processing capacity is most fully utilized and spare regional capacity is smallest.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor with DCS pre-positioning capability and precision kinetic delivery. Processing plants with significant NGL production are also accessible to a Tier 2 actor, given the voluntary physical security standards.

Modality: Cyber-kinetic combined arms. DCS compromise suppresses high-pressure and cryogenic temperature alarms and inhibits ESD actuation, allowing process conditions to develop that the safety systems would normally prevent. Kinetic action targets the turbo-expander cold box and the deethanizer column simultaneously — destroying the NGL extraction capability and the ethane separation capability in a single coordinated strike. The amine unit is a secondary target at sour gas facilities, selected to maximize off-site H₂S consequence.

Target selection: The plant or plants in the processing corridor whose loss produces the greatest compound consequence — maximum gas supply curtailment upstream, maximum NGL production loss, and maximum petrochemical feedstock disruption downstream. In the Permian

Basin, a plant serving multiple upstream gathering systems that feeds a major NGL pipeline hub represents the optimal compound-consequence target.

Timing: Peak winter demand or peak agricultural propane demand season, when processing throughput is highest, spare regional capacity is lowest, and the downstream consequence of NGL curtailment — residential heating propane shortages, agricultural crop drying disruptions — is most immediately visible to affected populations.

Simultaneity: Coordinated attacks on two or more processing plants within the same corridor within a window that exhausts the corridor's spare processing capacity — forcing wellhead shut-ins across the affected production basin and NGL supply curtailment across the petrochemical feedstock system simultaneously.

3.3 Dual Design Basis Architecture — The Optimization-Safety Boundary

Consistent with the DBA-GAS Sector Framework (§5.4) and DBA-EN-SF (§4.2), a natural gas processing facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Gas processing plants present an AI governance challenge that is qualitatively different from any other facility class in the DBA-GAS series, and more acute than the analogous challenge in petroleum refining. The commercial optimization pressure is structural: NGL prices are quoted on real-time commodity exchanges, inlet gas composition changes continuously with production mix, and the marginal value of an additional percent of ethane recovery can be significant at scale. This creates continuous economic pressure for APC and RTO systems to push process conditions — cryogenic temperatures, turbo-expander inlet pressures, demethanizer bottoms temperatures — toward the limits that maximize NGL recovery. Those limits are also the limits at which safety systems are most likely to be tested.

The architectural consequence is that the Bright Line in gas processing must be drawn not only against adversarial manipulation but against the plant's own optimization logic. An APC system that autonomously adjusts the demethanizer bottoms temperature to maximize ethane recovery is a below-Bright-Line ML system operating within its trained envelope — until it pushes the bottoms temperature below the SIS setpoint for low-temperature ESD actuation, at which point the optimization output conflicts with the safety system design basis. The Design Basis requirement is that this boundary cannot be approached autonomously. The four AI governance interaction points below establish the architecture that enforces it.

APC and RTO systems for NGL recovery optimization. APC and RTO systems that continuously adjust turbo-expander inlet conditions, demethanizer operating parameters, and fractionation column profiles to maximize NGL recovery are ML systems operating below the

Bright Line when their set point recommendations remain within a defined safety margin of the SIS operating envelope. They are deterministic, formally testable within their training envelopes, and may operate autonomously within those bounds. The critical constraint — unique to processing plants in the DBA-GAS series — is that the APC operating envelope must be defined with explicit safety margin setbacks from each SIS trip point, and those setbacks must be enforced by architecture, not by procedure. An APC system whose operating envelope overlaps with the SIS trip zone is a design basis violation regardless of how the operator characterizes the overlap in its process safety management documentation. The AI Governance Design Basis must establish that APC envelope limits are reviewed and approved by the facility's process safety engineer at each model update cycle, that APC limit changes that reduce the margin to any SIS trip point require management of change review under the OSHA PSM standard, and that the APC system has no pathway to override or suppress a SIS trip signal regardless of the optimization recommendation.

Gen AI advisory tools for process optimization and scheduling. Gen AI advisory tools for processing plant optimization — tools that recommend feedstock routing, product mix decisions, plant turnaround scheduling, or NGL price optimization strategies — are non-deterministic, not formally testable, and operate above the Bright Line. A human Command Broker must be in the control loop for any process parameter change recommended by a Gen AI advisory tool. The processing plant context adds a dimension absent from most other facility classes: Gen AI advisory tools with access to real-time NGL price data and real-time process conditions can generate optimization recommendations that are economically attractive precisely because they push process conditions toward the safety system limits where the highest NGL recovery rates are achievable. The Command Broker requirement is therefore not merely a governance formality — it is the mechanism by which the plant's commercial optimization logic is prevented from autonomously eroding the safety margins that the Design Basis requires. A Gen AI tool whose recommendations are translated into APC set point changes without Command Broker review violates the architecture regardless of whether the individual recommended change, in isolation, appears safe. The pattern of successive Gen AI recommendations, each individually within bounds, can produce a systematic drift toward operating conditions that are collectively outside the Design Basis envelope.

SIS Gen AI advisory tools and management of change. Safety instrumented systems at processing plants manage a wider range of process hazards than at any other facility class in the DBA-GAS series: high-pressure ESD, cryogenic low-temperature ESD, H₂S detection and evacuation, fired heater management, and fractionation column pressure relief coordination. Gen AI advisory tools for SIS bypass authorization, SIS logic modification, and proof-test scheduling are above the Bright Line and require Command Broker authorization — specifically, the review and affirmative approval of the qualified process safety engineer responsible for the SIS, not merely operator acknowledgment. The management of change process under OSHA PSM provides the procedural framework; the AI Governance Design Basis establishes that the PSM MOC process is the Command Broker mechanism for SIS parameter changes, and that it applies

to SIS parameter changes recommended by Gen AI tools in the same manner as SIS parameter changes initiated by any other source.

Fired heater management and the optimization-safety boundary at the fired equipment layer. Processing plants operate multiple fired heaters — reboilers for amine regeneration and fractionation columns, feed heaters for molecular sieve regeneration, and utility heaters. Fired heater management Gen AI advisory tools are above the Bright Line and require Command Broker authorization, consistent with the requirement established for petroleum refineries in DBA-OIL-FC1 §3.3. The processing plant context adds the interaction between fired heater optimization and the cryogenic process: a fired reboiler operating at high heat release to maximize fractionation column throughput interacts with the upstream cryogenic train through the process energy balance. A Gen AI tool that recommends increased reboiler duty to improve column separation at the same time that the APC system is pushing the turbo-expander toward its cold-temperature limit is generating recommendations whose compound effect on the process energy balance cannot be evaluated without human review of the system-level interaction. The Command Broker requirement for fired heater Gen AI tools must therefore extend to the evaluation of the recommendation in the context of the concurrent APC operating point. This system-level review cannot be delegated to either the APC system or the fired heater advisory tool independently.

4. Design Basis Accident Scenarios

4.1 Scenario A: Coordinated Cyber-Kinetic Attack on a Major Processing Complex

4.1.1 Initiating Event

Coordinated cyber-kinetic attack against a major turbo-expander processing complex in the Permian Basin processing corridor, serving multiple upstream gathering systems and producing significant NGL volumes for delivery to the Waha Hub and the West Texas LPG pipeline system. The cyber component, having achieved DCS access through a multi-stage intrusion campaign exploiting the plant's remote monitoring connection over the preceding months, initiates the coordinated manipulation of the turbo-expander inlet conditions and the demethanizer bottoms temperature — driving the cryogenic train toward conditions that create cold-temperature embrittlement risk in the cold box piping while simultaneously suppressing the low-temperature alarms that would trigger operator intervention.

The DCS manipulation is timed to execute at peak plant throughput — maximum inlet gas flow, maximum cryogenic operating depth, maximum NGL recovery — when the safety margins are thinnest, and the kinetic consequence of a piping failure is greatest. The kinetic component executes within the alarm-suppression window: precision UAS strikes against the cold box

structure and the deethanizer column reboiler, initiating a cryogenic hydrocarbon release and fire at the cold box and a high-pressure propane release at the deethanizer reboiler.

4.1.2 Progression Sequence

T+0: Cryogenic release at the cold box and propane release at the deethanizer reboiler ignite simultaneously. DCS alarm suppression prevents immediate operator detection of the severity. ESD is partially actuated; suppression of the low-temperature alarm delays the automatic isolation of the cryogenic train.

T+0 to T+30 minutes: Cold box fire intensifying as cryogenic inventory feeds the release. Deethanizer reboiler fire producing a propane vapor cloud in the fractionation area. Plant emergency shutdown initiated by operator manual action as the fire extent becomes visible. Inlet compression tripped. Upstream gathering system pressures are rising as the plant intake stops. Mutual aid fire response requested.

T+30 minutes to T+4 hours: Cold box fire requires burnout — foam application is ineffective on a cryogenic fire. Deethanizer fire is suppressed after the propane inventory depletes. Structural damage assessment: cold box support structure thermally damaged, deethanizer reboiler and lower column section destroyed. Upstream gathering system operators are initiating wellhead shut-ins as the plant intake is unavailable. DCS forensic investigation initiated.

T+4 to T+72 hours: Fires extinguished. Physical damage assessment: cold box requires full replacement — estimated 8 to 16 months. Deethanizer column lower section requires replacement — estimated 4 to 8 months. NGL production from the plant has ceased; upstream wellheads shut in across multiple gathering systems. The downstream ethylene cracker feedstock supply from this plant has been curtailed. Propane supply to the West Texas LPG system has been reduced.

T+72 hours onward: Cold box procurement initiated — long-lead cryogenic heat exchange equipment with limited global manufacturing capacity. DCS forensic investigation continuing. Wellhead shut-ins upstream are reducing Permian Basin gas production by the plant's inlet capacity. NGL pipeline operators are rerouting available supply from unaffected plants. Ethylene cracker operators assessing alternative ethane sourcing — NGL pipeline rerouting can partially offset but not fully replace the affected plant's ethane production within the petrochemical feedstock system.

4.1.3 Consequence Envelope

Gas supply curtailment. Upstream wellhead shut-ins reduce Permian Basin gas production by the plant's full inlet capacity — typically 200 to 800 MMcf/d for a major processing complex — for the duration of the cold box replacement. Associated gas production from Permian oil wells is curtailed simultaneously; operators face the choice of flaring (where permitted and economically preferable to shut-in) or reducing oil production to the rate at which associated gas

can be handled by remaining processing capacity. The compound consequence — gas supply reduction plus oil production curtailment — extends the economic impact of the attack significantly beyond the gas supply chain alone.

NGL supply curtailment. NGL production from the affected plant ceases for the cold box replacement duration of 8 to 16 months. The ethane, propane, butane, and natural gasoline volumes from the plant are removed from the NGL pipeline system simultaneously. Ethylene crackers receiving ethane from the affected plant's NGL pipeline system must identify alternative ethane sources or reduce throughput; those that have converted from naphtha to ethane feedstock have limited ability to swing back without operational adjustment time measured in weeks to months.

Recovery timeline. The governing constraint is cold box procurement and replacement. Cryogenic cold boxes for large turbo-expander plants are engineered to the specific plant's gas composition, flow rate, and NGL recovery depth. There are a limited number of manufacturers globally with the capability to fabricate and test large-scale cryogenic cold boxes; lead times of 8 to 16 months under normal supply chain conditions are the governing constraint. The deethanizer column replacement (4 to 8 months) runs in parallel. Still, it does not guarantee a restart — the plant can operate in a limited mode without the deethanizer if the cold box is repaired first, though at reduced NGL recovery.

4.2 Scenario B: Sour Gas Processing Plant Attack — H₂S Mass Casualty

4.2.1 Initiating Event

Targeted kinetic attack against the amine treating unit and thermal oxidizer at a major sour gas processing plant in the Midcontinent or Gulf Coast processing corridor, designed to achieve a large H₂S release. The attack employs the same dual-consequence logic as the HF alkylation unit attack in DBA-OIL-FC1: the H₂S processing unit is targeted not primarily for its contribution to gas supply output — amine unit damage can be repaired in weeks — but for its off-site consequence potential. H₂S is acutely toxic at concentrations far below those achievable in a large processing plant release; the 2006 Bandar Imam Khomeini petrochemical complex H₂S release in Iran, and the pattern of fatal sour gas well blowouts in producing regions, establish the lethal-cloud consequence envelope for large H₂S releases from processing infrastructure.

4.2.2 H₂S Release Consequence Envelope

On-site consequences. Fatalities and injuries among plant personnel in the amine treating and sulfur recovery areas. Emergency evacuation of the plant. Site emergency response focused on H₂S cloud tracking and personnel accountability rather than fire suppression, as H₂S is not primarily a fire hazard at the concentrations produced by an amine unit failure.

Off-site community consequences. Downwind H₂S vapor cloud extending potentially one to several miles from the release point under stable atmospheric conditions, depending on release

rate, H₂S concentration, and atmospheric stability class. The immediately dangerous to life and health (IDLH) concentration for H₂S is 100 ppm; lethal concentrations (above 500 ppm) can persist for extended distances from a large point release before dilution reduces concentrations below the IDLH threshold. Emergency shelter-in-place or evacuation orders for the surrounding community. Many Midcontinent and Gulf Coast processing plants are located in rural areas with limited emergency response resources, and the evacuation logistics for a large H₂S release affecting a dispersed rural population present challenges qualitatively different from the urban refinery scenario in DBA-OIL-FC1.

Regulatory consequence. OSHA and EPA investigations. Potential operational restrictions on sour gas processing operations pending safety review. The political pressure generated by a visible mass-casualty event at a gas processing plant — particularly if associated with the broader military-action context that motivated the attack — creates demand for reactive regulatory responses analogous to those identified for HF alkylation in DBA-OIL-FC1 §7.2.

4.3 Scenario C: Processing Corridor Cluster Attack — The Permian Simultaneity Trap

4.3.1 Initiating Event

Coordinated attacks against three or four major processing plants in the Permian Basin processing corridor within a 72-hour window, each employing the Scenario A cyber-kinetic modality independently. The simultaneity is the attack's defining characteristic: the Permian Basin's spare processing capacity — which provides resilience against a single-plant outage — is exhausted when three or four major plants are offline simultaneously. Upstream producers face a choice between flaring associated gas and curtailing oil production; many will choose curtailment to preserve reservoir integrity and comply with flaring restrictions.

4.3.2 The Permian Processing Bottleneck Applied

The 2018-2019 Permian Basin processing bottleneck — where rapid production growth outpaced processing capacity additions — demonstrated the mechanism: when processing capacity is constrained, upstream oil production is curtailed because associated gas has nowhere to go. The DBA-MA scenario applies deliberate destruction to produce the same bottleneck, but with two differences from the market-driven version. First, the destruction is simultaneous and exceeds the available spare capacity by design. Second, the cold box replacement timeline — 8 to 16 months — is far longer than the 6 to 12 months required to bring new greenfield processing capacity online, meaning the attack-induced bottleneck cannot be resolved by the same construction-acceleration that resolved the market-driven bottleneck.

The compound consequence of simultaneous Permian Basin processing plant destruction extends to the global crude oil market: the Permian Basin is the largest and fastest-growing crude oil-producing basin in the United States, accounting for approximately 6 million barrels per day of

production. Forced curtailment of Permian oil production due to associated gas processing constraints reduces U.S. crude oil production by an amount significant enough to affect global oil price benchmarks. This strategic consequence extends beyond the domestic gas supply chain and into the international energy security environment.

4.3.3 Consequence Envelope: Processing Corridor Attack

Gas supply impact. Simultaneous outage of three or four major Permian Basin processing plants removes 600 MMcf/d to over 2 Bcf/d of processing capacity from the corridor. Upstream wellhead shut-ins cascade across multiple gathering systems. Permian Basin gas supply to downstream markets — the Waha Hub, El Paso Natural Gas, Transwestern — is curtailed, affecting power generation in the Southwest and gas supply to Mexico under existing cross-border contracts.

Oil production curtailment. Forced curtailment of associated gas production forces oil production reductions at affected wells. A major Permian processing corridor attack can produce oil production curtailment of 500,000 to 1,500,000 barrels per day, depending on the flaring headroom available to affected operators. This is the most significant single-event domestic crude oil production reduction scenario in the DBA-OIL and DBA-GAS series.

NGL and petrochemical cascade. Simultaneous curtailment of NGL production from multiple major processing plants reduces ethane supply to Gulf Coast ethylene crackers, propane supply to residential and agricultural markets, and butane supply to gasoline blending. The petrochemical feedstock disruption triggers supply allocation decisions across the chemical sector within weeks of the processing corridor attack.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in U.S. gas processing infrastructure or specific processing plant control architectures. Anomalous DCS network activity at processing plants: unusual process historian queries, anomalous vendor remote access sessions, and lateral movement between corporate IT and process control networks inconsistent with normal operations. Physical surveillance of processing plant perimeters in producing corridor areas. Procurement inquiries for cold box specifications, cryogenic piping specifications, or processing plant engineering data. Unusual aerial activity over processing plant facilities or approach corridors.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized DCS access at any processing plant. Simultaneous DCS anomalies at two or more processing plants within the same corridor. Confirmed unauthorized aerial UAS in processing plant airspace. Anomalous cryogenic temperature trends are inconsistent with normal process variation at a turbo-expander plant. Simultaneous security incidents at two or more processing plants within a producing corridor within 72 hours. Confirmed kinetic event at any natural gas processing facility.

5.2 Decision Authority Framework

Facility level: Plant operator authority to initiate emergency shutdown, isolate process units, and activate fire suppression and evacuation systems. TSA and CISA notification for confirmed or suspected cyber incidents affecting DCS systems. EPA and OSHA notification for process safety incidents. Local emergency management notification for H₂S releases with potential off-site consequences.

Federal coordination: CISA for DCS cyber incident coordination. DOE ESF-12 for supply disruption response affecting the transmission system gas supply. EPA for significant H₂S releases with community consequences. FBI and DOD for events with confirmed or strongly suspected state-actor attribution.

Upstream and pipeline operator coordination: Gathering system operators and transmission pipeline operators whose systems connect to affected processing plants must be notified immediately to manage upstream wellhead pressure and downstream supply allocation. Pre-established notification protocols between processing plant operators and connected gathering and transmission operators are a current gap in the emergency response framework, parallel to the terminal-pipeline notification gap identified in DBA-OIL-FC3.

5.3 Federal Coordination Thresholds

Confirmed kinetic strike at any major natural gas processing facility. Confirmed DCS compromise with suspected nation-state attribution at any gas processing plant. H₂S release affecting or threatening to affect the surrounding community above a defined concentration threshold. Simultaneous security incidents at two or more processing plants within a producing corridor within 72 hours. Curtailment of gas processing capacity affecting transmission system supply above a defined threshold. Oil production curtailment resulting from associated gas processing constraints above a defined threshold.

6. Recovery Architecture

6.1 Processing Plant Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency shutdown and fire suppression. H₂S consequence management and community notification for sour gas events. DCS forensic

investigation initiation. Gathering system operator notification and upstream wellhead shut-in coordination. Transmission operator notification for downstream supply allocation. Cold box and fractionation column damage assessment and procurement inquiry initiation — procurement timeline assessment is Phase 1, not Phase 2, because it governs the recovery critical path. Alternative NGL supply routing assessment for downstream petrochemical consumers.

Phase 2 — Short-term (72 hours to 8 weeks). DCS forensic investigation continuing. Physical damage assessment complete. Cold box and column procurement is underway. Upstream gathering system coordination for partial plant restart using undamaged train capacity where available. NGL pipeline operators are formalizing alternative supply arrangements. Ethylene cracker operators are activating alternative feedstock sourcing or throughput reduction plans.

Phase 3 — Medium-term (8 weeks to recovery). Cold box fabrication and delivery. Cryogenic train reconstruction and commissioning. Fractionation column section replacement in parallel. Regulatory inspection and startup authorization. Sequential restart of processing trains as each is repaired and commissioned. Supply chain monitoring for adversary interdiction of cryogenic equipment and specialty column internals.

6.2 Critical Recovery Constraints

Cryogenic cold box procurement. The governing recovery constraint for Scenario A and C events. Large cryogenic cold boxes for turbo-expander plants are engineered to plant-specific specifications and fabricated by a small number of global manufacturers — primarily in Germany, Japan, and the United States. Lead times of 8 to 16 months reflect both the engineering and fabrication time and the limited manufacturing capacity globally available for large-scale cryogenic heat exchange equipment. No strategic reserve of processing plant cryogenic equipment exists. Pre-positioned spare cold box modules — or advance procurement agreements with manufacturers for priority fabrication slots — would be the highest-consequence single mitigation for the Scenario A and C recovery timeline.

Fractionation column section replacement. Deethanizer, depropanizer, and debutanizer column sections destroyed in a targeted attack require procurement of replacement column shells, trays, or packing, and reboiler and condenser systems. Lead times of 4 to 8 months run in parallel with cold box procurement, but do not gate restart if the cold box is the longer item. Column section procurement is subject to the same supply chain interdiction risk as cold box procurement.

DCS reconstitution. Processing plant DCS forensic investigation and reconstitution must be completed before automated process operations can resume. The minimum 4 to 6 week timeline is consistent with the refinery and terminal DCS reconstitution timelines established in DBA-OIL-FC1 and DBA-OIL-FC3. The processing plant context adds the requirement that the APC operating envelope be re-validated against the SIS trip points following reconstitution — the

reconstituted DCS must be re-certified to operate within the safety margins established by the Design Basis before NGL optimization can resume.

Associated gas flaring and oil production curtailment management. For Permian Basin cluster attacks, the recovery architecture must address the upstream oil production curtailment as an active consequence management problem, not a passive consequence. Accelerated permitting for emergency flaring operations — to allow upstream producers to continue producing oil while processing capacity is restored — requires coordinated regulatory action at the state and federal levels. The Texas Railroad Commission and EPA flaring authorization processes are not designed for emergency acceleration at the scale required by a Scenario C event.

7. Gaps and Recommendations

7.1 Regulatory Gap: Absence of Mandatory Cybersecurity Standards for Processing Plant DCS

TSA Pipeline Security Directives apply to gas pipeline operators. Processing plant DCS systems — the primary cyber attack surface for the Scenario A bounding event — are not covered by these directives unless the processing plant is owned by a pipeline operator subject to the directives. Independent processing plant operators whose DCS architecture includes the same IT-OT boundary vulnerabilities as pipeline SCADA systems have no mandatory cybersecurity framework. TSA, in coordination with CISA, should evaluate extending the pipeline cybersecurity directive framework to cover processing plant DCS systems at facilities whose outage would produce supply disruptions above a defined threshold.

7.2 APC Operating Envelope Safety Margin — Regulatory Gap

OSHA PSM requires process hazard analysis and management of change for processing plant operations. It does not explicitly require that APC operating envelopes be defined with explicit safety margin setbacks from SIS trip points, nor does it require that APC envelope modifications go through the PSM MOC process. This gap allows APC systems to be configured — through incremental optimization adjustments, each individually below the PSM MOC trigger threshold — to operate at progressively reduced margins to SIS trip points without formal process safety review. OSHA should issue guidance clarifying that APC operating envelope modifications that reduce the margin to any SIS trip point constitute a process change subject to PSM MOC requirements, regardless of the magnitude of any individual adjustment.

7.3 Cryogenic Equipment Strategic Reserve

The cold box procurement constraint is the governing recovery timeline for major turbo-expander plant attacks and has no strategic reserve analog in any federal program. The electric sector's STEP program and the OIL sector's discussion of a pump unit strategic spare program

provide the analytical model. DOE, in coordination with PHMSA and the natural gas processing industry, should assess the feasibility of pre-positioned cryogenic cold box modules — either as manufacturer-held inventory under government reserve agreements or as operator-held spares at the highest-consequence processing corridor locations. The Permian Basin processing complex, where plant concentration and oil production association produce the most consequential single-corridor attack scenario, is the highest-priority target for such a program.

7.4 Associated Gas Flaring Emergency Authorization Framework

A Scenario C Permian Basin processing corridor attack forces upstream oil producers to choose between shutting in oil production and flaring associated gas at rates that exceed normal EPA and Texas Railroad Commission flaring authorization levels. No emergency authorization framework exists for accelerated flaring permitting at the scale required by a multi-plant processing outage. The Texas Railroad Commission and EPA should develop an emergency flaring authorization protocol — triggered by a declared energy supply emergency — that allows upstream producers to continue oil production under elevated flaring rates for a defined period while emergency processing capacity is restored. The protocol must include air quality monitoring requirements and a defined sunset condition tied to processing capacity restoration milestones.

7.5 NGL Supply Chain Emergency Response Framework

The NGL consequence chain — processing plant curtailment propagating through NGL pipelines to ethylene crackers, residential propane suppliers, and agricultural propane users — crosses sector boundaries that no existing emergency response framework spans. DOE ESF-12 governs energy supply emergency response, but has historically focused on electric power and natural gas transmission. The NGL and petrochemical feedstock consequences of a major processing plant attack are not addressed in any federal emergency response plan. DOE, in coordination with the American Chemistry Council and the National Propane Gas Association, should develop an NGL supply emergency response annex to ESF-12 that identifies the consequence thresholds, federal coordination mechanisms, and alternative supply options for each major NGL product stream.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the natural gas processing facility class. Issued under DBA-GAS Sector Framework v1.4. Three scenarios were developed: Scenario A (coordinated cyber-kinetic attack on a major turbo-expander processing complex — cold box replacement as governing recovery constraint), Scenario B (sour gas processing plant attack —

			H2S mass casualty consequence envelope), Scenario C (Permian Basin processing corridor cluster attack — simultaneity trap with associated oil production curtailment). Dual Design Basis architecture established with four processing-specific AI governance interaction points, incorporating the series-first recognition that the Bright Line in continuous-process optimization facilities must constrain the plant's own commercial optimization logic as well as adversarial manipulation. Regulatory gaps identified: processing plant DCS cybersecurity, APC operating envelope safety margin, cryogenic equipment strategic reserve, associated gas flaring emergency authorization, and NGL supply chain emergency response framework.
--	--	--	--